Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/09/2005
PI Name:	Goree, John Ph.D.		
Project Title:	Optically-Excited Waves in 3D Dusty Plasmas		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	john-goree@uiowa.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	319-335-1843
Organization Name:	University of Iowa		
PI Address 1:	Department of Physics and Astronomy		
PI Address 2:	210 Van Allen Hall		
PI Web Page:			
City:	Iowa City	**State:**	IA
Zip Code:	52242-1403	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	04/01/2002	**End Date:**	03/31/2006
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Rogers, Richard	**Contact Phone:**	216-433-6512
Contact Email:	Richard.B.Rogers@grc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NAG3-2771		
Performance Goal No.:			
Performance Goal Text:			
	Original Abstract: Experiments will be carried out in an interdisciplinary study of waves (i.e., phonons) in 3D Coulomb lattices. The experimental system is a suspension of charged polymer microspheres in a gas-discharge plasma. This so-called "dusty plasma" is a macroscopic system where the microspheres arrange themselves, due to their mutual repulsion, in a lattice. Shear waves will be excited in the lattice using the radiation pressure of a laser beam. New Strategic Abstract: During the final two years of this grant, the research goals are shifted to emphasize strategic goals related to NASA's planned manned space flight missions to the Moon and Mars. Countermeasures for dust contamination in Advanced EVA will be the theme. Experiments will be performed to test materials and investigate processes related to dust adhesion on surfaces and the use of plasma exposure to remove dust from those surfaces. Original Objective: We will prepare for future flight experiments, which will have two tasks: * We will study the		

Task Description:	dispersion relation (i.e., phonon spectrum) of sinusoidal waves and the propagation of pulsed waves. These experiments will be performed with plane waves. * We will study the formation of Mach cones, which are the V-shaped wakes or shocks created by a supersonic disturbance. In this experiment, a moving laser spot is the supersonic disturbance. New Strategic Objective: Preliminary experiments will be carried out to investigate adhesion properties of JSC-1 Lunar Simulant dust to EVA suits, and the effectiveness of plasmas in removing that dust. Original description: * Need for microgravity: These experiments are an extension of 2D experiments performed in the laboratory, under gravity conditions. Sedimentation due to gravity causes the microspheres to settle rapidly into a 2D layer that is levitated by the strong electric field near an electrode in a plasma. By eliminating gravity, it is possible to fill a 3D volume with the suspension. * Use of a novel experimental system: Our experiments will exploit the low damping of a dusty plasma as compared to colloidal suspensions. We will excite waves using a manipulation laser, which applies a radiation pressure on the particles. Using a dusty plasma allows us to measure particle motion in-situ using direct imaging with video microscopy. * Our scientific effort and our team are interdisciplinary. New Strategic Task Significance: Advanced EVA has been identified as one of the four critical technologies for NASA's New Vision for Space Exploration. The EVA System includes EVA suits and airlocks, for which dust contamination is a risk. According to Lara Kearney, JSC's manager of Advanced EVA, 13 challenges of present-day technologies have been identified, and two of these include dust: the present-day EVA suit protection from dust intrusion is inadequate, and dust on airlocks is an unsolved risk. Our project will include studies for low-TRL technology needs related to dust on the Moon and Mars. “Iowa Dust Mitigation Scheme”: We are performing experiments at The University of Iowa to test methods of cleaning space suits. In our “Iowa Dust Mitigation Scheme,” the crew of a lunar habitat would use two steps to clean their suits after returning to the habitat from an EVA. Firstly, they will use air to blow dust off the suits during airlock pressurization, and secondly, after egress from the suit, they will use plasma to electrostatically clean the residual dust. These two steps have the following advantages: (1) they can both be done in the airlock, with dust collecting at the bottom of the airlock, thereby isolating it from entering the habitat; (2) they will not require any water, which will be a precious resource; (3) they do not involve any frictional rubbing of the suit’s fabric, which would reduce the suit’s lifetime. 1st step of the “Iowa Dust Mitigation Scheme”: We will exploit the availability of air flow during pressurization of the airlock. We propose that air should not be introduced into the airlock diffusely, but instead it should be introduced through a hose that the astronaut will direct onto the spacesuit, while still wearing the suit. This will blow off a large fraction of the dust, which would collect at the bottom of the airlock where it can be emptied periodically. 2nd step of the “Iowa Dust Mitigation Scheme”: The second step will exploit the availability of vacuum conditions in an airlock. After the astronaut has removed the space suit and egressed, the suit will hang in the airlock. The airlock will then be partially evacuated, with a gas pressure of approximately 1 mBar. Radio-frequency electrical power will be applied, ionizing the gas to make plasma that will be exposed to the airlock’s contents, including the suit. Electrostatic effects will remove dust from the suit. This method was first introduced by T.E. Sheridan et al., J. Geophys. Res. 97, 2935 (1992), and it has been used in the semiconductor industry for cleaning particulates from silicon wafers. While our scheme is intended to be used in a lunar Habitat, it could also be adapted to a habitat on Mars, which will also require airlocks. The electrostatic effects that will remove dust from the suit will be the primary focus of this research. When a particle rests on a surface, it is held in place by a van-der-Waals attraction, but this adhesion can be overcome by several mechanisms. One such mechanism for the Moon and Mars environments is the application of electrostatic forces. Scientific results from this project will also find other applications: (1) One issue is cleaning solar panels on Mars. Dust storms on Mars will gradually cover the panels, reducing their efficiency, unless they are cleaned. Because Mars has a very low-pressure atmosphere, it may be possible to use radio-frequency power to ionize the Martian atmosphere immediately above a solar panel, and thereby clean it. (2) Another issue is gaining an understanding of dust particle leviation and transport on the Moon. The Moon, unlike Mars, has no atmosphere. Instead, it is immersed in interplanetary plasma, which consists of free electrons and protons at a low number density. The Moon disturbs this plasma, so that there is an electric field perpendicular to the Moon’s surface. This electric field applies a vertical force to any electrically-charged fine particle. Particles are known to become charged and levitated on the Moon. This was discovered in Apollo missions, where a thin cloud of dust was seen floating above the lunar horizon. Particles can be charged either photoelectrically by exposure to solar UV, or by collection of free electrons and ions from the surrounding plasma. This can lead to levitation, if the particle can first overcome van-der-Waals adhesion to surfaces such as a rock, a photovoltaic cell, or another dust particle.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
	Original Task: Experiments will be carried out in an interdisciplinary study of waves (i.e., phonons) in 3D Coulomb lattices. The experimental system is a suspension of charged polymer microspheres in a gas-discharge plasma. This so-called "dusty plasma" is a macroscopic system where the microspheres arrange themselves, due to their mutual repulsion, in a lattice. Shear waves will be excited in the lattice using the radiation pressure of a laser beam. Accomplishments for original task: During the reporting period, we have published papers in scientific journals reporting our experimental results. We were also featured in the cover story of Physics Today, which is a news publication of the American Institute of Physics, with a circulation of 120,000, including public libraries around the country. New Strategic Task: During the final two years of this grant, the research goals are shifted to emphasize strategic goals related to NASA's planned manned space flight missions to the Moon and Mars. Countermeasures for dust contamination in Advanced EVA will be the theme. Experiments will be performed to test materials and investigate processes related to

Task Progress:	dust adhesion on surfaces and the use of plasma exposure to remove dust from those surfaces. Accomplishments for new strategic task: During the reporting period, we built hardware for testing cleaning schemes for EVA space suit materials tests, and we have performed these tests. The schemes involve using airflow or plasma to remove dust from surfaces. We have performed these tests using a swatch of EVA suit fabric, JSC-1 lunar stimulant dust, and a video camera to image dust flying off the fabric during the cleaning. This was accomplished ahead of schedule, as compared to the milestone. We reported preliminary results at the "Conference-Workshop on Strategic Research to Enable NASA's Exploration Missions," held in Cleveland in June 2004. Further results will be reported at a NASA workshop in Golden, CO in May 2005. Advanced EVA test results: This project is motivated by the Bioastronautics Critical Path Roadmap, Research & Technology question #37c, "How do we protect against planetary surface dust through suit and airlock system design?" This question is rated priority 1 in the Roadmap. It is also motivated by the AIM (Advanced Integrated Matrix) Dust Assessment, http://lstworkshop.arc.nasa.gov/ We have performed tests in our laboratory to demonstrate the feasibility of two methods of removing dust from EVA space suits. These tests were done to meet a milestone for July 2005, and they were completed ahead of schedule, in March 2005. We've gone beyond our original milestone by testing not just one but two schemes. In our tests we used a vacuum chamber as a proxy for an airlock. We used JSC-1 lunar simulant that we applied to the exterior surface of the space suit fabric. Below, I describe briefly the two schemes, along with our tests. We performed this work in cooperation with Sandy Wagner of JSC's Advanced EVA office. 1. Airflow scheme. We have demonstrated the effectiveness of a simple scheme. We don't know for sure if it's a new idea, but we've never heard of it previously in the context of cleaning space suits. In this scheme, air that is vented into the airlock during pressurization would be directed through a hose, held by the astronaut, to blow dust off the suit. This will be done while he/she is wearing the suit. Air comes out of a nozzle and blows dust off the suit. The dust would collect at the bottom of the airlock due to gravity, or in a filter. This scheme has the advantages of simplicity and of making use of what is already available: the supply of air that is vented into the airlock to bring it up to the habitat's pressure. We have recorded videos showing dust particles flying off the surface of the space suit fabric when a nozzle is placed nearby, during venting to atmospheric pressure. We have also identified some design issues involving airflow that would require attention to implement this scheme. 2. Plasma scheme. We have demonstrated the scheme we originally proposed for dust removal from a space suit due to exposure to a plasma. The plasma is made by applying electrical power to a plasma source when the airlock is under vacuum conditions at less than 1 milliBar vacuum. In this scheme, the spacesuit would be cleaned after the astronaut has egressed it, and after the airlock is mostly evacuated again. We have recorded videos showing dust particles flying off the fabric when exposed to the plasma. We have also identified some design issues that will require further work, such as selecting and testing an efficient type of plasma source. We propose that these two schemes can both be used in an airlock in a two-step process: first the airflow from the hose to remove most of the dust, and second the plasma cleaning to remove residual dust. Relationship to non-NASA projects: • The original "Waves in 3D Dusty Plasmas" task will lead to experiments that can be performed on ISS using a European Space Agency (ESA) facility, IMPACT/IMPF. This ESA facility is now in a Phase-B planning stage. ESA is planning on providing use of the IMPACT facility to the PI for this project, as a result of the PI's top-rated proposal "Waves in 3D Dusty Plasmas", which was originally submitted in response to the IAO (International Announcement of Opportunity). From ESA's perspective, this project has foreign co-I's, principally in Germany. • During the reporting period, the PI received a letter from NASA HQ informing the PI that an additional project, "Self-structuring in Dusty Plasmas," was approved for funding. This project has foreign co-I's. The entire project is led by Professor Alexander Piel, of the University of Kiel, in close collaboration with the PI, who will visit the PI for one month, beginning 6 April, 2005.

Bibliography Type:	Description: (Last Updated: 01/23/2025)
Articles in Peer-reviewed Journals	V. Nosenko, K. Avinash, J. Goree and B. Liu "Nonlinear Interaction of Compressional Waves in a 2D Dusty Plasma Crystal" Vol. 92, 085001-1– 085001-4, 2004 , Jan-2004
Articles in Peer-reviewed Journals	B. Liu, K. Avinash and J. Goree "Characterization of Interparticle Potential from Lattice Structure" Physical Review E, Vol. 69, 036410, 2004 , Jan-2004
Articles in Peer-reviewed Journals	Robert L. Merlino and John A. Goree "Dusty Plasmas in Industry, the Laboratory and Space" Physics Today, Vol. 57, 32-38 , Jul-2004
Articles in Peer-reviewed Journals	V. Nosenko and J. Goree "Shear Flows and Shear Viscosity in a Two-Dimensional Yukawa System (Dusty" Physical Review Letters, Vol. 93, pp. 155004-1 - 155004-4 , Jan-2004
Articles in Peer-reviewed Journals	B. Liu and J. Goree "Natural Phonons in a One-Dimensional Yukawa Chain: Dusty Plasma Experiment and" Submitted to Physical Review E Accepted , Mar-2005